ST. ELIGIUS HOSPITAL SYSTEMS MODERNIZATION

Systems Analysis Presentation for the Board of Directors

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MGMT 480 — Systems Analysis & Design

THE CRISIS: LEGACY SYSTEM FAILURE

Outdated System Architecture

- St. Eligius currently operates on a 30-year-old legacy procedural system
- Data and system logic are stored separately across multiple systems

Data Fragmentation

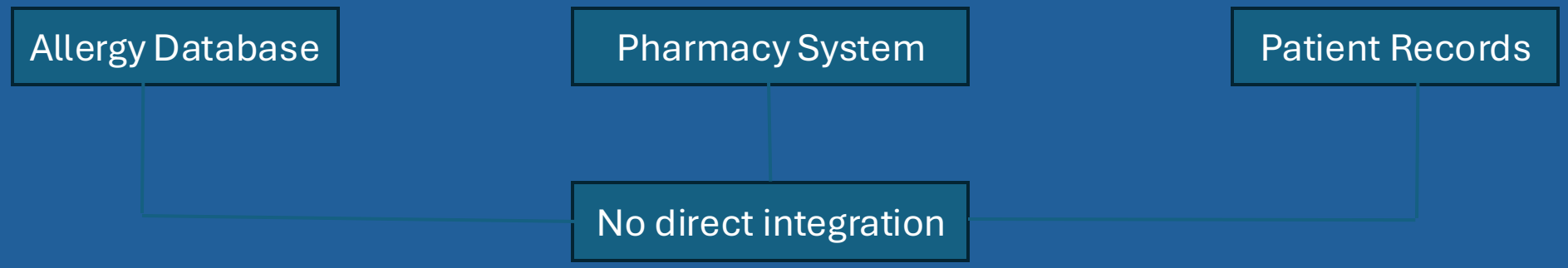
- Critical patient information exists in isolated data silos
- Clinical data does not always synchronize with system decision logic

Clinical Safety Risk

- Allergy information may not connect with pharmacy dispensing systems
- This disconnect has already led to near-miss medication errors

Operational Inefficiency

- Staff spend approximately 30% of their shifts manually reconciling patient information
- Time spent navigating multiple systems delays patient care



**Disconnected systems create clinical risk and operational inefficiency.*

THE SYSTEMS ANALYST: THE BRIDGE

Systems Analyst = translator between clinical staff and developers

- Connects hospital workflows with technical system design

Clinical World (Stakeholders)

- Doctors
- Nurses
- Pharmacists
- Laboratory staff

Technical World (Implementation)

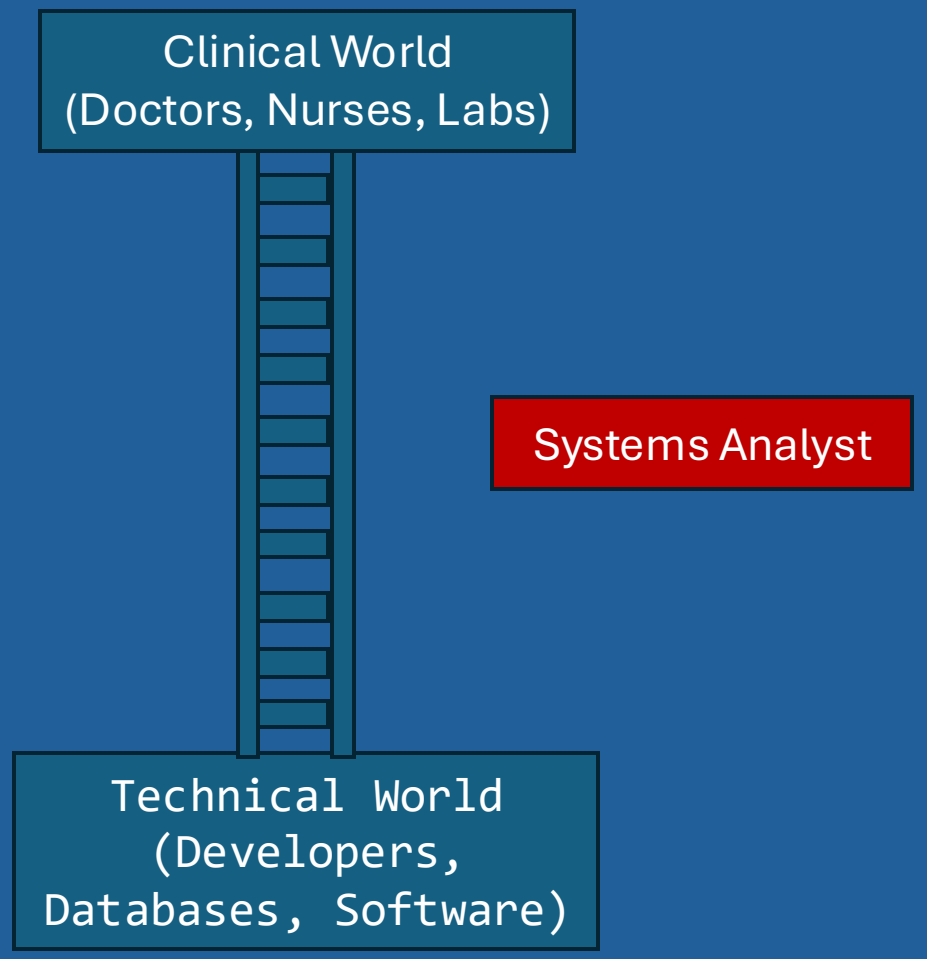
- Software developers
- Databases
- System architecture

Core Responsibility

- Gather clinical requirements
- Convert workflows into logical system models

Why This Role Matters

- Ensures the system supports real hospital operations
- Prevents technology that does not align with clinical workflows



**The Systems Analyst ensures medical workflows and system architecture remain aligned.*

CLASSES: THE SYSTEM BLUEPRINT

What is a Class?

- A class is a blueprint used to create objects in a system
- It defines what data exists and what actions can occur

St. Eligius System Blueprint

- The Patient_Record class defines the structure for all patient records
- Every patient in the hospital system is created from this blueprint

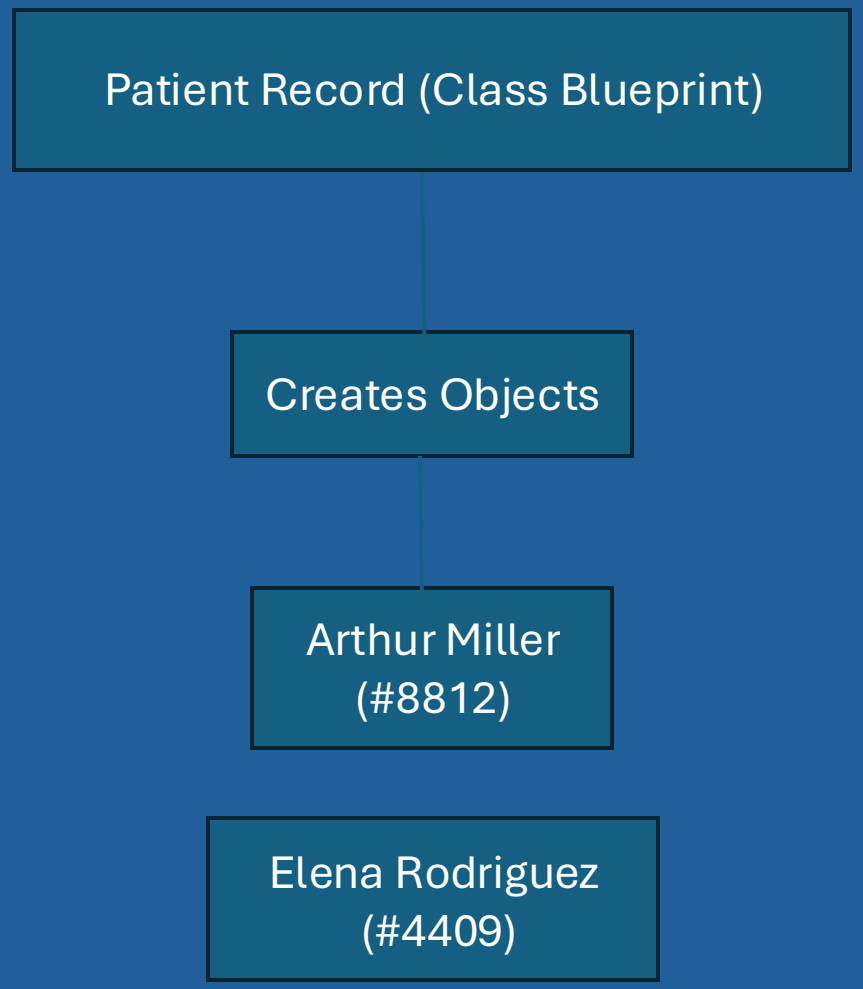
What the Class Defines

The Patient_Record class specifies:

- Patient information stored in the system
- Clinical operations that can be performed
- Rules that protect patient data

Why This Matters

- Ensures consistent structure for every patient record
- Reduces data fragmentation across hospital systems
- Provides the foundation for a safer and more integrated system



**The Patient Record class acts as the blueprint for all patient records.*

OBJECTS: REAL PATIENTS IN THE SYSTEM

What is an Object?

- An object is a specific instance created from a class
- Each object represents a unique entity in the system

Class Used in the System: Patient_Record

Patient Objects in the St. Eligius System

Object 1

- Patient: Arthur Miller
- Patient ID: #8812

Object 2

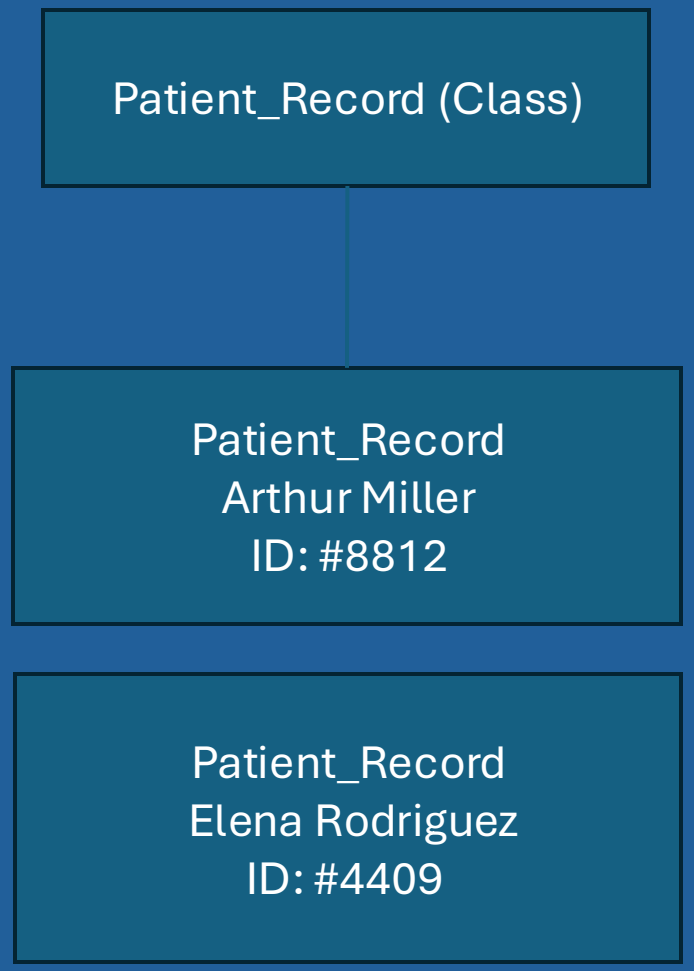
- Patient: Elena Rodriguez
- Patient ID: #4409

Key Idea

- Each patient object stores its own data independently
- The system can manage thousands of patient records using the same blueprint

Why This Matters

- Ensures every patient record is consistent and organized
- Allows the hospital system to track and manage individual patients accurately



**Objects are individual patient records created from the Patient_Record class.*

ATTRIBUTES: PATIENT DATA

What Are Attributes?

- Attributes represent the data stored inside an object
- They describe the state of the patient record

Attributes in the Patient_Record Class

- Blood_Type
- Active_Allergies
- Insurance_Status
- Current_Vital_Signs

Example Patient Record

- Arthur Miller (#8812)
- Blood Type: O+
- Active Allergies: Penicillin
- Insurance Status: Active
- Current Vital Signs: Stable

Why This Matters

- Critical patient information is stored securely within the object
- Medical staff can access accurate patient data instantly
- Reduces the risk of data mismatches across systems

Patient_Record Object

Arthur Miller (#8812)

Blood_Type: O+

Active_Allergies: Penicillin

Insurance_Status: Active

Current_Vital_Signs: Stable

**Attributes store the critical clinical data associated with each patient.*

METHODS: CLINICAL ACTIONS

What Are Methods?

- Methods are functions that an object performs
- They define the behavior of the patient record

Methods in the Patient_Record Class

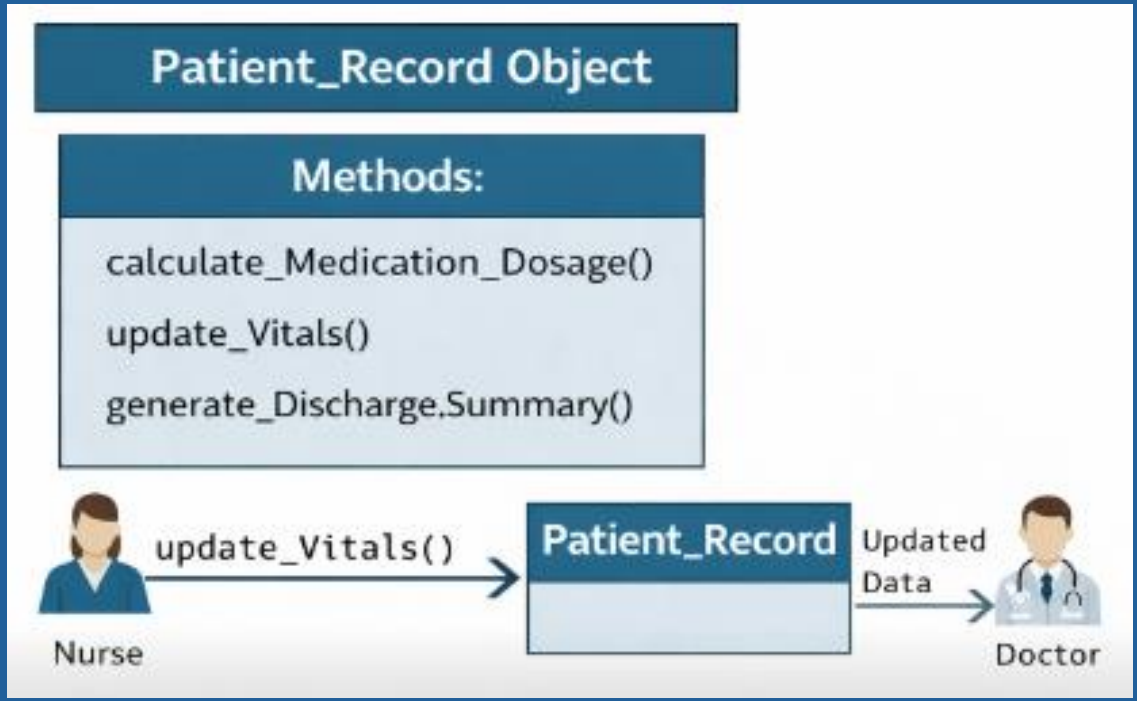
- calculate_Medication_Dosage()
- update_Vitals()
- generate_Discharge_Summary()

Example Clinical Workflow

- Nurse records new vital signs
- update_Vitals() updates the patient record
- Doctor immediately sees updated patient data

Why This Matters

- Automates important clinical processes
- Reduces manual data entry
- Ensures accurate and up-to-date patient information



**Methods allow the patient record to actively manage clinical information.*

MESSAGES: SYSTEM COMMUNICATION

What Are Messages?

- Messages allow objects to communicate with each other
- One object requests information or actions from another object

Example in the St. Eligius System

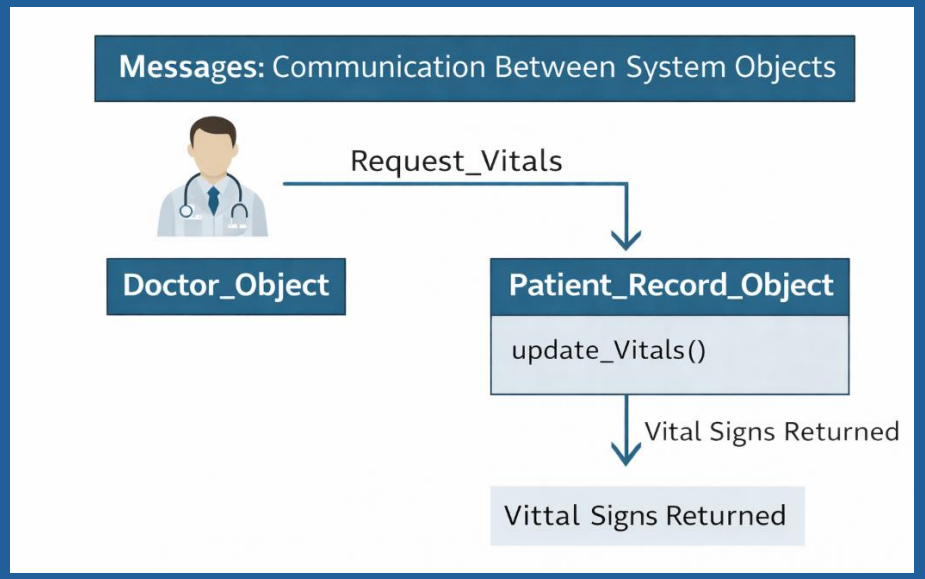
- A doctor requests patient vitals.
- Doctor_Object → Request_Vitals → Patient_Record_Object

How the Interaction Works

- Doctor requests patient vitals
- A Request_Vitals message is sent to the patient record
- The patient record uses its internal method to retrieve the data
- The system returns the updated vital signs to the doctor

Why This Matters

- Enables real-time clinical information access
- Reduces time spent searching for patient data
- Improves speed and accuracy of medical decisions



**Messages allow system components to communicate and deliver clinical information instantly.*

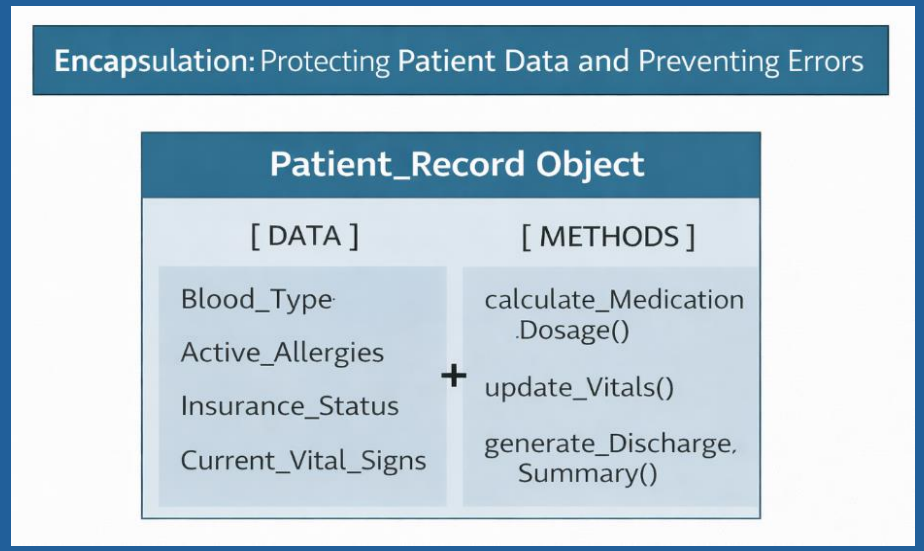
ENCAPSULATION: PROTECTING PATIENT DATA AND PREVENTING ERRORS

What is Encapsulation?

- Encapsulation means bundling data and behavior inside the same object
- The object controls how its data is accessed and used

In the St. Eligius System

- The Patient_Record object contains both:
- Patient data (allergies, vitals, insurance)
- System logic (methods that act on the data)



How Encapsulation Prevents Errors

Example:

- Allergy data is stored inside the patient record
- Medication dosage calculations automatically check allergy data

Result:

- Prevents medication conflicts
- Reduces risk of clinical mistakes
- Ensures safer patient care

Operational Benefit

- Eliminates fragmented data across systems
- Improves reliability of clinical information

**Encapsulation bundles patient data and system logic together to prevent medical errors.*

SCALABILITY: PREPARING ST. ELIGIUS FOR FUTURE EXPANSION

What is Scalability?

- Scalability means the system can grow and adapt without major redesign
- New departments and technologies can be added without disrupting the existing system

How Object-Oriented Design Enables Scalability

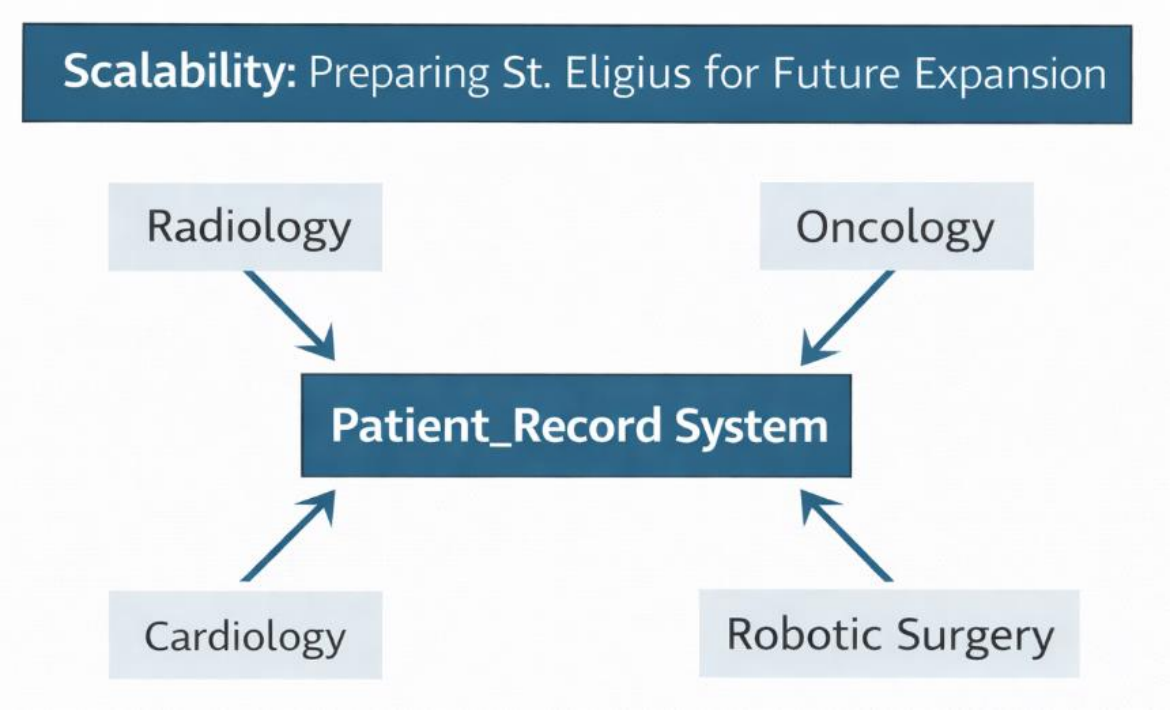
- The Patient_Record class serves as a universal blueprint
- New hospital systems can interact with the same patient record structure

Example Future Expansions

- Cardiology Department
- Oncology Department
- Robotic Surgery Unit
- Advanced Diagnostic Labs

Why This Matters

- Supports long-term hospital growth
- Reduces future IT redevelopment costs
- Ensures patient data remains consistent across all departments



**Object-oriented architecture allows new hospital departments to integrate easily with the patient record system.*

THE VALUE PROPOSITION: WHY THIS SYSTEM MATTERS

Patient Safety

- Integrated patient records reduce the risk of medication conflicts
- Allergy information is automatically checked during treatment decisions

Operational Efficiency

- Eliminates manual reconciliation between multiple systems
- Allows doctors and nurses to access patient data instantly

System Reliability

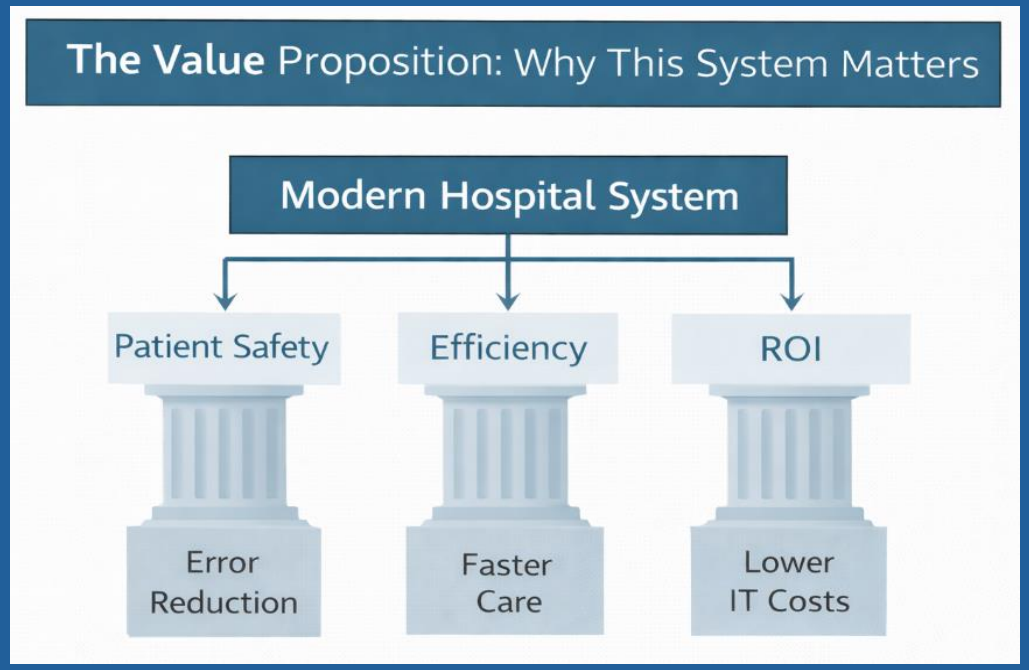
- Object-oriented design creates consistent and accurate patient records
- Reduces system errors caused by fragmented data

Financial Return on Investment (ROI)

- Less staff time spent managing disconnected systems
- Reduced medical error costs
- Scalable infrastructure lowers long-term IT expenses

Strategic Outcome

- St. Eligius gains a safer, faster, and more scalable hospital information system



**Object-oriented architecture improves safety, efficiency, and financial performance.*

QUESTION & ANSWERS

St. Eligius Systems Modernization

Object-Oriented Analysis enables:

- Safer patient care
- Faster clinical workflows
- Integrated hospital systems
- Scalable infrastructure for future expansion

THANK YOU FOR YOUR TIME AND CONSIDERATION!

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MGMT 480 – Systems Analysis & Design

REFERENCES

- ChatGPT (Used to create some of the visual images)
- *St. Eligius Systems Modernization Project Brief.* MGMT 480 – Systems Analysis & Design, course handout.